

Dormitory Management System

Evaluation Report

Branch: Achieve-all-functional-requirements

Module: Advanced Web Development — University of Hildesheim

Report Date: 2026-03-15

EVALUATION CRITERIA

Metric	Options	Description
Achievement	Fully Achieved · Partially Achieved · Not Achieved	Whether the requirement is met in the current implementation
Remove	Should Remove · Can Remove · Shouldn't Remove	Whether the implementation introduces overkill or scope creep beyond the requirement

1. FUNCTIONAL REQUIREMENTS — MUST

ID	Requirement	Achievement	Justification
R01	Student Registration	Fully Achieved	Full frontend RegisterComponent with all required fields (name, email, password, phone, student ID, course, university). Calls AuthService.register() ! POST /api/register. Linked from login page via goToRegister().
R02	Student Auth & Login	Fully Achieved	/login page, AuthService, POST /api/login, JWT storage, redirect to /student-dashboard. Works end-to-end.
R02a	Admin Auth & Login	Fully Achieved	Same login flow with role-based redirect to /dashboard. JWT payload includes role and dormitory_id.
R03	Online Application Form	Fully Achieved	Apply section backed by POST /api/applications. Auto-detects NEW vs EXTENSION type. Application history table shown.
R04	Student Profile	Fully Achieved	Profile section with GET/PUT /api/profile. Personal and academic info editable and saved to DB. Live dormitory/room data from JOIN.
R05	Admin Dashboard	Fully Achieved	Full dashboard with live stats, activity feed, rooms, residents, payments, complaints, reports, and settings — all from live API.
R06	Application Review	Fully Achieved	Residents section with three sub-tabs (New, Extension, Termination). GET /api/applications role-filtered.
R07	Application Acceptance	Fully Achieved	Two-step flow: inline form ! acceptDialog modal ! transactional room assignment + automatic contract PDF generation.
R07a	Application Rejection	Fully Achieved	rejectDialog modal: remarks ! confirm ! result phases. PATCH /api/applications/:id/status with REJECTED and optional remarks.
R08	File Upload (App Docs)	Fully Achieved	POST /api/applications/:id/files. Up to 10 files (PDF/JPG/PNG, 5 MB). Client-side size validation added to all upload handlers.
R09	Digital Contract Creation	Fully Achieved	Contract auto-created on application acceptance. PDFKit contract PDF saved to /uploads/contracts/.
R10	Upload Signed Contract	Fully Achieved	POST /api/contracts/sign. 10 MB server limit. Client-side size validation (10 MB) before upload. One-time-only enforced server-side.
R11	Download Contract	Fully Achieved	GET /api/contracts/download streams PDF to student. Download button activates after contract PDF is generated.

ID	Requirement	Achievement	Justification
R12	Upload Payment Receipt	Fully Achieved	POST /api/payments with Multer (5 MB server limit). Client-side size validation (5 MB). Receipt path saved to rent_payments.
R13	Store Payment Records	Fully Achieved	POST /api/payments stores to DB. GET /api/payments retrieves payment history with verification status badges.
R14	Download Payments as PDF	Fully Achieved	GET /api/reports/:id/download streams PDFKit-generated PDF report with payment summary, complaint list, and occupancy stats.
R15	Contract Extension	Fully Achieved	Students submit EXTENSION application. Admin reviews in dedicated Extension Requests sub-tab using the same Accept flow.
R16	Contract Termination	Fully Achieved	Two paths: (1) student-initiated termination request sub-tab and (2) admin-initiated adminTerminateDialog modal. Both mark contract TERMINATED and free the room.
R17	Backend Long-Running Report	Fully Achieved	POST /api/reports uses setImmediate() for non-blocking PDF generation. generatePdfReport() updates progress table at 25/50/75/100% milestones.
R18	Progress Status Info	Fully Achieved	GET /api/reports/:id/progress returns { percentage, status }. Frontend polls every 600ms; stops on COMPLETED/CANCELLED/FAILED.
R19	Visual Progress Indicator	Fully Achieved	reportModal with live progress bar, status label, Cancel button, and Download button (activates on COMPLETED). Bar reflects true backend progress.
R20	Complaint Submission	Fully Achieved	Complaint form triggers complaintDialog confirmation before API call. History list with SUBMITTED/IN_PROGRESS/RESOLVED badges.
R21	Complaint Management	Fully Achieved	Admin requests section. SUBMITTED != IN_PROGRESS != RESOLVED workflow via PATCH /api/complaints/:id/status.
R22	Dialog: Complaint Confirm	Fully Achieved	complaintDialog modal: confirm phase shows preview; result phase shows success/error. API called only after student confirms.
R23	Dialog: App Confirm	Fully Achieved	acceptDialog (contract details summary + confirm) and rejectDialog (remarks != confirm != result) modal overlays for admin.
R24	Angular Child Routes	Fully Achieved	Child routes under /dashboard/* and /student-dashboard/*. URL updates per section (e.g. /dashboard/rooms). activeNav synced from router via NavigationEnd events.
R25	RBAC: Students	Fully Achieved	authGuard + roleGuard applied to /student-dashboard via canActivate. Server-side requireRole(STUDENT) on all student endpoints.
R25a	RBAC: Admins	Fully Achieved	authGuard + roleGuard applied to /dashboard via canActivate. Server-side requireRole(ADMIN) on all admin endpoints.

2. FUNCTIONAL REQUIREMENTS — SHOULD

ID	Requirement	Achievement	Justification
R26	Application Status Overview	Fully Achieved	Overview cards and apply section show application status (PENDING/ACCEPTED/REJECTED) from GET /api/applications.
R26a	Contract Status Overview	Fully Achieved	Contract section: status badge, dates, rent, due day, overdue warning, termination request status. All live from GET /api/contracts.
R27	Admin Pending Apps Overview	Fully Achieved	Pending Apps count card from GET /api/stats. Activity feed highlights recent applications. Quick-action navigates to residents section.
R28	Complaint Status Visibility	Fully Achieved	Student complaints section shows each complaint with SUBMITTED/IN_PROGRESS/RESOLVED status badge from GET /api/complaints.

ID	Requirement	Achievement	Justification
R29	Filter Payments by Month	Partially Achieved	Month dropdown exists for payment submission form (selecting which month to pay). No separate filter control exists for the payment history list.
R30	Document Validation	Fully Achieved	Server: Multer enforces type (PDF/JPG/PNG) and size limits (5 MB / 10 MB). Client: all four upload handlers validate file.size before assigning. accept attribute provides browser-level type filtering.
R31	Notification Messages	Fully Achieved	Inline success/error alerts after API calls. Dialog result phases. SSE real-time bell notifications and toast pop-ups on both dashboards.

3. FUNCTIONAL REQUIREMENTS — CAN

ID	Requirement	Achievement	Justification
R32	Search Functionality	Not Achieved	No search or filter input exists in the admin dashboard for tenants, applications, payments, or complaints.
R33	Data Export	Not Achieved	No CSV or Excel export functionality. Admin reports are PDF-based; no tabular data export provided.
R34	Student Profile Update	Fully Achieved	GET/PUT /api/profile fully implemented. Student can edit personal and academic info; saved to DB.

4. TECHNICAL REQUIREMENTS

ID	Requirement	Achievement	Justification
TR01	Angular Router Navigation	Fully Achieved	Angular 21 with child routes /dashboard/* and /student-dashboard/*. authGuard + roleGuard applied. activeNav synced from ActivatedRoute via NavigationEnd.
TR02	Backend Technology	Fully Achieved	Express.js + Node.js + MySQL2 + JWT + bcrypt + Multer + PDFKit. 40+ REST endpoints with full auth and role middleware.
TR03	Automated Tests (Karma)	Not Achieved	Project uses Vitest, not Karma — toolchain mismatch. Only 2 trivial tests exist in app.spec.ts (component creation + title text). No service, component, or integration tests.
TR04	Test Coverage 60%+	Not Achieved	Effective coverage near 0%. No E2E tests. Two existing tests check only that the root component renders. Significant gap requiring immediate attention.
TR05	VM Delivery 2026-03-17	Not Assessed	Deadline not yet reached at report date. VM preparation and documentation packaging status not evaluated.

5. SUMMARY SCORECARD

Category	Fully Achieved	Partially Achieved	Not Achieved
Must (R01–R25a)	26	0	0
Should (R26–R31)	5	1 (R29)	0
Can (R32–R34)	1 (R34)	0	2 (R32, R33)
Technical (TR01–TR05)	2 (TR01, TR02)	0	2 + 1 N/A
TOTAL	34	1	4 + 1 N/A

Progress Since Previous Branch (Payments-due-contract)

- R01 — Student registration: full frontend RegisterComponent added
- R24 — Angular child routes for all dashboard sections implemented
- R25 — authGuard + roleGuard applied to /student-dashboard
- R25a — authGuard + roleGuard applied to /dashboard
- R30 — Client-side file size validation in all four upload handlers
- TR01 — Angular router-based navigation now fully implemented

Remaining Gaps

ID	Status	Description
R29	Partially Achieved	Month dropdown is for payment submission, not for filtering the payment history list
TR03	Not Achieved	Tests use Vitest, not Karma; only 2 trivial tests with no real coverage
TR04	Not Achieved	Effective coverage near 0%; 60%+ required including E2E tests
TR05	Not Assessed	VM delivery deadline: 2026-03-17